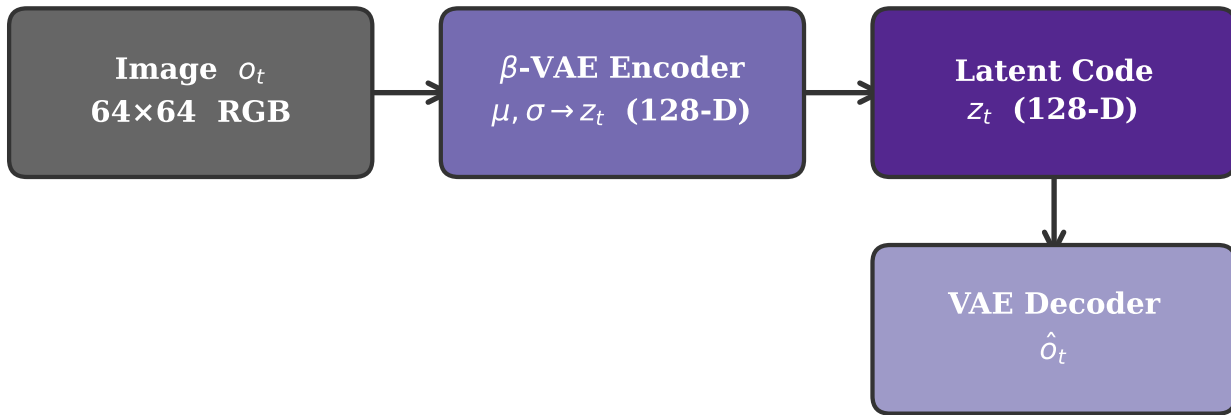


SAC + β -VAE — Phase 1: Visual Representation

*Pre-trained on 30 episodes of random exploration.
Encoder weights frozen before SAC training begins.*



$$\mathcal{L}_{\text{VAE}} = \|o_t - \hat{o}_t\|^2 + \beta D_{\text{KL}}(q\|p), \quad \beta = 0.1$$